

◆ AWS SOLUTIONS ARCHITECT ASSOCIATE (SAA-C03)

STUDENT LEARNING NOTES

MZ-UCA Cloud Academy

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◆ WHAT IS A SOLUTIONS ARCHITECT?

Student Notes

A Solutions Architect is responsible for designing cloud solutions that meet business and technical requirements. The role is not only about deploying resources but about ensuring that systems are secure, scalable, reliable, and cost-effective.

When a company moves to AWS, the Solutions Architect decides:

- Which AWS services should be used
- How those services should connect together
- How to keep the environment secure
- How to ensure the application remains available during failures
- How to optimize cloud costs

A Solutions Architect must always balance four important areas:

Security

Protect data, applications, and users.

Reliability

Ensure systems continue working even when components fail.

Performance

Ensure applications respond quickly and efficiently.

Cost Optimization

Ensure the business does not spend more than necessary.

Important Exam Insight

The AWS SAA exam is not a memorization exam.

AWS wants to know whether you can choose the best architectural solution for a given scenario.

When reading exam questions, always ask yourself:

"What is the business trying to achieve?"

◆ AWS WELL-ARCHITECTED FRAMEWORK

Student Notes

The AWS Well-Architected Framework is a collection of best practices that helps architects build secure, efficient, and reliable cloud environments.

AWS uses five pillars to evaluate architecture quality.

1. Operational Excellence

Operational Excellence focuses on running and managing systems effectively while continuously improving them.

Examples include:

- Monitoring systems using CloudWatch
- Tracking activities using CloudTrail
- Automating operations through Infrastructure as Code

Key Question:

"Can the organization operate and improve this environment efficiently?"

2. Security

Security focuses on protecting information, systems, and assets.

Examples include:

- IAM permissions
- Encryption using KMS
- Threat detection using GuardDuty

Key Question:

"How is access controlled and data protected?"

3. Reliability

Reliability focuses on ensuring systems continue functioning despite failures.

Examples include:

- Multi-AZ deployments
- Auto Scaling Groups
- Backup and recovery solutions

Key Question:

"Can the system recover from failure?"

4. Performance Efficiency

Performance Efficiency focuses on using resources efficiently as business needs change.

Examples include:

- CloudFront
- DynamoDB
- Elastic Load Balancing

Key Question:

"Can the system maintain performance as demand grows?"

5. Cost Optimization

Cost Optimization focuses on avoiding unnecessary expenses.

Examples include:

- Rightsizing resources
- Storage lifecycle policies
- Savings Plans

Key Question:

"Are we getting maximum value from our cloud spending?"

Exam Reminder

Whenever you see the phrase "AWS best practice," think about the Well-Architected Framework.

◆ **AMAZON VIRTUAL PRIVATE CLOUD (VPC)**

Student Notes

Amazon VPC allows you to create a private network inside AWS.

Think of a VPC as your company's private section within the AWS cloud.

Inside the VPC, you control:

- IP address ranges
 - Subnets
 - Routing
 - Security rules
 - Internet connectivity
-

Public Subnet

A public subnet contains resources that need direct internet access.

Examples:

- Web servers
- Public-facing applications
- Bastion hosts

Public subnets typically use an Internet Gateway.

Private Subnet

A private subnet contains resources that should not be accessible from the internet.

Examples:

- Databases
- Internal application servers
- Backend services

Private subnets provide a stronger security posture.

Internet Gateway (IGW)

An Internet Gateway enables communication between resources in a VPC and the internet.

Without an Internet Gateway, resources cannot communicate directly with the internet.

NAT Gateway

A NAT Gateway allows resources in private subnets to access the internet without exposing them to incoming internet traffic.

Example:

A private EC2 instance needs to download operating system updates.

The NAT Gateway allows outbound communication while maintaining security.

Exam Tip

If AWS asks:

"How can private servers access the internet while remaining private?"

The answer is usually:

NAT Gateway

◆ HIGH AVAILABILITY AND FAULT TOLERANCE

Student Notes

High Availability means ensuring systems remain available even when failures occur.

AWS achieves this through Availability Zones (AZs).

An Availability Zone is an isolated data center within an AWS Region.

Multi-AZ Design

Instead of placing all resources in one Availability Zone, distribute them across multiple Availability Zones.

Benefits:

- Reduced downtime
 - Improved resilience
 - Better fault tolerance
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Auto Scaling

Auto Scaling automatically adjusts the number of servers based on demand.

When demand increases:

- New servers are launched automatically

When demand decreases:

- Unnecessary servers are removed

Benefits:

- Improved performance
 - Reduced costs
 - Better user experience
-

Load Balancing

A Load Balancer distributes incoming traffic across multiple servers.

Benefits:

- Prevents server overload
 - Improves availability
 - Supports scaling
-

Exam Reminder

If AWS mentions:

- Traffic spikes
- Automatic capacity adjustments
- Maintaining performance during growth

Think:

Auto Scaling Group

◆ **AWS STORAGE SERVICES**

Student Notes

AWS offers three primary storage models.

Understanding the differences is essential for the SAA exam.

Amazon S3

Amazon S3 is object storage.

It stores files as objects inside buckets.

Common use cases:

- Backups
- Static websites
- Images and videos
- Data lakes

One of the most tested AWS services.

Key Feature:

11 nines of durability (99.999999999%).

Amazon EBS

Amazon Elastic Block Store (EBS) provides block-level storage.

EBS volumes are attached directly to EC2 instances.

Common use cases:

- Operating systems
- Databases
- Enterprise applications

Think:

"EBS behaves like a hard drive."

Amazon EFS

Amazon Elastic File System (EFS) provides shared file storage.

Multiple EC2 instances can access the same file system simultaneously.

Common use cases:

- Shared application data
- Linux file systems
- Web server farms

Think:

"EFS behaves like a shared network drive."

Quick Memory Rule

S3 = Object Storage

EBS = Block Storage

EFS = Shared File Storage

◆ DATABASE ARCHITECTURE

Student Notes

AWS provides different database services depending on business requirements.

Amazon RDS

Amazon RDS is a managed relational database service.

Examples:

- MySQL
- PostgreSQL
- MariaDB
- SQL Server

Use RDS when relationships between data are important.

Examples:

- Banking systems
 - ERP systems
 - E-commerce platforms
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Amazon DynamoDB

DynamoDB is a managed NoSQL database.

Benefits:

- Extremely fast
- Highly scalable
- Serverless

Common use cases:

- Mobile applications
 - Gaming platforms
 - IoT workloads
-

Multi-AZ

Creates a standby database in another Availability Zone.

Purpose:

Improve availability.

Read Replica

Creates additional copies of a database for read operations.

Purpose:

Improve performance.

Easy Exam Rule

Multi-AZ = Availability

Read Replica = Performance

◆ IAM AND SECURITY

Student Notes

IAM (Identity and Access Management) controls access to AWS resources.

IAM answers three important questions:

Who can access AWS?

What can they access?

What actions can they perform?

Users

Represent individual people.

Example:

An employee accessing AWS.

Groups

Collections of users with similar permissions.

Example:

Developers Group

Administrators Group

Roles

Temporary permissions assumed when required.

Roles are heavily tested in the exam.

Examples:

- EC2 accessing S3
 - Lambda accessing DynamoDB
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Policies

Policies define permissions using JSON documents.

Policies determine:

- Allowed actions
 - Denied actions
 - Specific resources
-

Principle of Least Privilege

Users should receive only the permissions required to perform their job.

Benefits:

- Improved security
 - Reduced risk
 - Better governance
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Exam Reminder

If AWS asks for the most secure solution:

Choose the option that follows:

Least Privilege

◆ **DISASTER RECOVERY**

Student Notes

Disaster Recovery focuses on preparing for failures and restoring operations quickly.

Recovery Time Objective (RTO)

How quickly systems must recover.

Lower RTO means faster recovery.

Recovery Point Objective (RPO)

How much data loss is acceptable.

Lower RPO means less data loss.

Backup and Restore

Cheapest option.

Recovery takes longer.

Pilot Light

Critical systems remain running.

Other resources are launched during recovery.

Warm Standby

A smaller version of the environment remains active.

Recovery is faster.

Active-Active

Multiple fully operational environments run simultaneously.

Fastest recovery.

Highest cost.

Exam Rule

Lower RTO and RPO usually mean higher costs.

◆ **COST OPTIMIZATION**

Student Notes

Cost Optimization is a major focus area for Solutions Architects.

The goal is to meet business requirements while minimizing unnecessary spending.

On-Demand Instances

Pay only for what you use.

Best for:

- Short-term workloads
 - Unpredictable workloads
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Reserved Instances

Commit to using resources for one or three years.

Benefits:

- Significant discounts
 - Predictable costs
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Savings Plans

Flexible pricing model that provides discounts for committed usage.

Frequently recommended by AWS.

Spot Instances

Use spare AWS capacity at very low prices.

Can be interrupted at any time.

Best for:

- Batch processing
 - Testing environments
 - Big data workloads
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Exam Reminder

Flexible workload + Lowest cost

Usually = Spot Instances

◆ FINAL SAA EXAM STRATEGY

Student Notes

When solving exam questions:

Step 1

Identify the primary requirement.

Is the question about:

- Security?
 - Availability?
 - Performance?
 - Cost?
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Step 2

Look for AWS-managed services.

AWS generally prefers managed services because they reduce operational overhead.

Step 3

Eliminate weak answers.

Remove options that:

- Increase operational complexity
 - Reduce security
 - Cost more than necessary
 - Do not fully solve the problem
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Step 4

Choose the solution that best satisfies the business requirement.

Remember:

The correct AWS answer is often the solution that provides:

- High Availability
- Strong Security
- Operational Simplicity

- Cost Efficiency

◆ GOLDEN SAA EXAM FORMULA

Whenever you are unsure during the exam, think:

Managed Services + High Availability + Least Privilege + Cost Optimization = The Most Likely Correct Answer

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"A Solutions Architect designs systems that scale. A Governance Architect designs systems that remain controlled while they scale."